SP4CD: A Hierarchical Stripe Patch-Based Method for Change Detection in Remote Sensing Images

Zhilin Qu, Lianghao Xu, Zehua Chen, *Member, IEEE*, Mingzhe Li, and Jiajie Shi

***Abstract*—Change detection (CD) is a fundamental task in remote sensing imagery, playing a pivotal role in diverse applications, such as urban planning, environmental monitoring, and natural disaster assessment. However, most existing methods focus on processing entire images, with limited exploration of cropping images into strip patches and performing operations within these patches. These methods suffer from a waste of computational cost and memory resources in many areas. In this article, we present a novel paradigm for the CD task and introduce a lightweight yet powerful framework, termed SP4CD. Specifically, SP4CD employs a Siamese backbone as a shared feature extractor to capture discriminative and hierarchical representations from bi-temporal remote sensing imagery. Furthermore, an interactive fusion mechanism is devised to integrate the extracted representations across spatial and channel dimensions, thereby enhancing cross-temporal feature interaction. To this end, we design a patch-wise intertemporal channel integration (PICI) module. By introducing a patch-swapping strategy, PICI integrates dual-temporal feature channels in a nearly parameter-free manner, achieving adaptive channel enhancement while effectively alleviating misclassification errors. In addition, we explore an intertemporal synergistic patch scanning (ISPS) module. It refines spatial representations through 2-D selective scan (SS2D) of patches, optimizing detail in changing regions to reduce false negatives. Moreover, two complementary scanning modes (parallel and series modes) are designed based on guidance sequences derived from distinct feature representations, enabling adaptive spatial exploration from multiple perspectives. Extensive experiments conducted on four challenging benchmark datasets verify that the proposed SP4CD achieves competitive or superior performance across multiple evaluation metrics, confirming its effectiveness and generalization capability. Our source code is available at: https://github.com/ZehuaChenLab/SP4CD**

***Index Terms*—Bi-temporal interaction, change detection (CD), Mamba, remote sensing, state-space model (SSM), strip patch.**

I. Introduction

CHANGE detection (CD) is a fundamental and significant research task in the field of remote sensing, aimed at analyzing the variability of images associated with specific regions captured in different time intervals [1]. CD plays a pivotal role in a wide range of real-world applications, including urban expansion monitoring [2], natural disaster assessment [3], land cover/use change analysis [4], and environmental protection [5].

Over the past few decades, a large number of point-wise classification methods [6], [7], [8] have been proposed to solve CD tasks. Although these machine learning-based methods can achieve considerable performance, their performance is highly dependent on decision classifiers and threshold settings [9]. The widespread availability of high-resolution satellite imagery has led to greater heterogeneity within surface regions, posing challenges to traditional methods to reliably identify areas of change.

The emergence of deep learning has provided a fresh perspective on CD tasks, and methods based on deep learning have achieved remarkable development by automatically learning discriminative representations from bi-temporal images. Convolutional neural network (CNN)-based methods [10], [11], [12] have made significant progress due to their ability to capture hierarchical spatial features through local receptive fields and weight sharing, enabling end-to-end optimization without relying on handcrafted descriptors. Siamese architectures and encoder–decoder networks [13], [14], [15] have further enhanced feature alignment and localization accuracy, leading to significant performance gains over traditional approaches. Although CNN-based methods have shown promising performance in CD, their intrinsic reliance on local receptive fields limits their ability to capture long-range contextual dependencies. This constraint hinders the accurate modeling of complex spatial structures present in remote sensing imagery, which can reduce the robustness of detection in challenging scenarios.

Recently, an increasing number of Transformer-based models have been introduced into CD. The rapid advancement of visual Transformers [16], [17] has made Transformer-based methods one of the most mainstream architectures. These methods [18], [19], [20] leverage self-attention mechanisms to model long-range dependencies and global contextual relationships between multitemporal observations, thus effectively addressing the disadvantage of CNN's limited receptive fields. However, the self-attention operation scales quadratically with the input size, resulting in substantial computational and memory overhead. Such inefficiency becomes particularly problematic for dense prediction tasks like CD. In recent times, Mamba [21] architecture has introduced an input-dependent selective attention mechanism based on the structured state-space sequence model (S4) [22], demonstrating significant

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Zhilin Qu, Lianghao Xu, Zehua Chen, and Mingzhe Li are with the College of Artificial Intelligence, Taiyuan University of Technology, Taiyuan 030024, China (e-mail: 2023520742@link.tyut.edu.cn; 2024510457@link.tyut.edu.cn; chenzehua@tyut.edu.cn; limingzhe111341@163.com).

Jiajie Shi is with the College of Computer Science and Technology (Big Data), Taiyuan University of Technology, Taiyuan 030024, China (e-mail: 19722753035@163.com).

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potential for modeling long sequences. Jiao et al. [23] proposed a 2-D selective scan (SS2D) and introduced it to images. Based on this, Mamba architecture is widely applied in CD [24], [25]. Furthermore, some methods [26], [27], [28] based on spiking neural networks (SNNs) have achieved remarkable results in the field of remote sensing due to their low power consumption and energy efficiency.

However, most of the methods often overlook the distribution patterns of the areas changed. In addition, while global attention mechanisms can model long-range dependencies, they incur significant computational overhead and tend to reduce sensitivity to local details, leading to decreased detection accuracy. Based on this, patch-based strategies [29], [30], [31] have been widely explored in CD to effectively address these issues. They further process the image at the patch level by converting the entire image into patches. Feature extraction is performed using patch pairs, and the order of patches is rearranged to generate diverse long sequences for sequence modeling. These methods effectively reduce the computational overhead associated with modules designed for the entire image. Despite these advances, most of the existing methods rely on a square patch, and there is little exploration of methods based on a strip patch.

In this article, we present a novel perspective for addressing CD tasks. Unlike methods based on CNN and Attention mechanisms, we conduct prior analysis and statistical assessment of the data distribution. Feature extraction and semantic interaction processes are implemented in a strip patch rather than in the entire image, thereby reducing the excessive computational overhead. Unlike patch-based methods, we control the shape of patches as horizontal and vertical strips. This allows a strip patch spanning the image to preserve target structures to a significant extent. Unlike traditional modified scanning approaches of Mamba or adding local detail branch models to Mamba, we apply scanning to our strip patch, enabling selective scanning within the patches. The key contributions of our method are summarized as follows.

1) We propose a novel binary CD network after analyzing the data distribution named SP4CD, which employs a Siamese architecture to acquire accurate semantic information and capture rich bi-temporal features in a strip patch.
2) We design a patch-wise intertemporal channel integration (PICI) module that takes a strip patch from cropped images as input. Through swapping, concatenating, dynamic selection, and decoupling, it enhances the fusion of temporal correlations across channel dimensions.
3) We explore an intertemporal synergistic patch scanning (ISPS) module. We employ a selective scan within a strip patch to capture spatial dependencies and enhance the temporal difference feature map to serve as the weight for spatial attention. In addition, we design two modes (parallel and series mode) based on the sequence of intertemporal guidance derived from different feature maps.
4) Qualitative and quantitative studies of four datasets demonstrate that our proposed SP4CD method is comparable to existing approaches. Extensive ablation studies validate its effectiveness. In the first line of the last paragraph of the introduction, we add a sentence: In this article, we present a novel perspective for addressing CD tasks. Its performance effect is shown in Fig. 1

II. Related Work

A. CNN-Based Methods

With the development and evolution of CNNs, their powerful hierarchical feature representation capabilities have led to widespread application in the early stages of the CD task. Daudt et al. [13] were the first to introduce the FCN method into CD. They accomplished the detection task by concatenating dual-temporal images along the channel dimension for feature extraction. After that, numerous representative Siamese network architectures were proposed. Fang et al. [14] adopted a densely connected Siamese architecture to process bi-temporal inputs in parallel, which facilitates efficient feature reuse. Huang et al. [11] developed a Siamese network, which enhances spatiotemporal differences and integrates multilevel features through interlevel fusion to improve the robustness of CD in complex scenes. In addition, a variety of methods for feature representation and enhancement continue to emerge. Zhang et al. [12] effectively enhanced the retention of local details and semantic consistency by integrating multiscale features from different layers of CNNs. By measuring the Euclidean distance between feature representations, Chen and Shi [10] produced difference maps that effectively suppressed pseudo changes. Lei et al. [32] improved feature extraction performance and accuracy through multiscale decoupled convolution combined with spatial–spectral feature collaboration.

Despite their notable success, the local nature of CNN receptive fields restricts these methods from capturing long-range relationships, particularly in CD tasks.

B. Transformer and Mamba-Based Methods

To overcome this limitation, researchers have incorporated Transformer architectures into CD. Unlike CNNs, Transformers can model long-range dependencies and global context through self-attention mechanisms, thereby obtaining more effective feature representations from dual-temporal images. Several studies have explored pure Transformer or hybrid CNN-Transformer frameworks to jointly extract local details and global structural variations, achieving superior accuracy and robustness. Chen et al. [33] pioneered the introduction of Transformers into the CD task and proposed the BIT. The approach enables effective context modeling through the sparsification of bi-temporal features into visual tokens. Bandara and Patel [34] explored ChangeFormer, a transformer-based Siamese network that combines a hierarchically structured transformer encoder with an MLP decoder to capture multiscale long-range features. Song et al. [35] proposed ACABFNet, a CNN-Transformer hybrid network that employs axial cross attention to efficiently capture local and global dependencies for accurate CD under limited data conditions. Noman et al. [19] designed ELGCNet, a Siamese encoder–decoder framework that integrates a local–global context aggregator using pooled-transpose attention and depth-wise convolutions to efficiently capture both global and local

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dependencies. Ren et al. [20] proposed an interactive dual-mode attention network to integrate bitemporal features, while employing supervised attention to enhance boundary detection.

Recently, state-space models (SSMs) have gained attention for visual tasks due to their linear computational complexity with respect to input sequence length, providing an efficient alternative for capturing long-range dependencies compared to standard Transformers, such as Mamba. With these approaches [23], [36] applying Mamba to visual tasks [37], similar methods can also be found in CD tasks. Zhao et al. [38] integrated an omnidirectional selective scan module to perform global modeling of image context from multiple directions, thereby capturing large-spatial features from different orientations for dense predictions. Chen et al. [39] employed Visual Mamba [23] as a backbone to capture global spatial context information and validated its generalization capabilities across multiple CD tasks. Zhang et al. [24] proposed CDMamba, which integrated global features from Mamba and local detailed clues via convolution to address the shortcomings of Mamba's suboptimal local modeling. Wu et al. [40] introduced a locally adaptive SSM for remote sensing CD, which enhances local feature perception while maintaining global awareness. The above methods have contributed to lightweight models, but there has been relatively little exploration of scanning within the strip patch.

C. Patch-Based Methods

To make CD tasks more cost-effective and resource-efficient, many studies have focused on the organizational form of data, transforming multitemporal images into patches for processing. Rahman et al. [29] proposed a Siamese network that operates on paired image patches rather than full scenes. Xue et al. [30] proposed LPCDNet, a lightweight patch-level CD network that efficiently identifies and removes unchanged patches, significantly reducing computational overhead. Liu et al. [41] cropped images from two temporal phases into patches, arranged these patches in multiple combinations, and selectively scanned various sequences to achieve effective spatial feature extraction and learn global contextual information. Chen et al. [31] developed a sample generation method by simply exchanging intra and interimage patches, which can train change detectors on single temporal images in an unsupervised manner. These methods effectively reduce computational overhead, facilitating the practical application of CD. Although significant progress has been made, most approaches use square patches, with limited investigation into strip patch-based methods.

III. Model Methods

A. Preliminaries

1) SSM and Mamba Architecture: SSMs are a type of system designed to model 1-D sequences or functions by leveraging the structure of linear time-invariant (LTI) systems. An SSM maps an input sequence $x(t) \in \mathbb{R}$ to an output sequence $y(t) \in \mathbb{R}$ via a hidden state $h(t) \in \mathbb{R}^N$, typically formulated as a linear ordinary differential equation (ODE)

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = Ch(t) \tag{2}$$

where $A \in \mathbb{R}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$ are system parameters. To make SSMs compatible with deep learning frameworks, continuous-time systems are often discretized using methods like zero-order hold (ZOH), resulting in

$$\bar{A} = \exp(\Delta A) \tag{3}$$

$$\bar{B} = A^{-1}(\exp(\Delta A) - I)(\Delta B). \tag{4}$$

Then, the system is updated in discrete time as

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t. \tag{5}$$

$$y_t = Ch_t. \tag{6}$$

Furthermore, (5) and (6) can also be mathematically transformed into an equivalent CNN form as follows:

$$\bar{K} = \left(C\bar{B}, C\overline{AB}, \ldots, C\bar{A}^{L-1}\bar{B}\right) \tag{7}$$

$$y = x \circledast \bar{K} \tag{8}$$

where L is the length of the input sequence, $\circledast$ denotes the convolution operation, and $\bar{K} \in \mathbb{R}^L$ is a structured convolution kernel.

Recent models, such as S4 [22] and Mamba [21], build on this foundation by introducing data-dependent parameters and GPU-efficient algorithms. Mamba dynamically learns input-dependent parameters $\Delta, \bar{B}, C$, allowing dynamic feature representation and flexible modeling. At the same time, the scanning mechanisms allow Mamba to enjoy the same advantages of parallel processing as (7) and (8), thus facilitating efficient training.

2) Fraction of Fully Retained Objects: In CD tasks, processing an entire image at once often incurs substantial computational cost. Consequently, it is common practice to partition large images into smaller patches to reduce the input size of models. However, such partitioning may fragment originally intact change targets across multiple patches, compromising their spatial integrity and potentially degrading detection accuracy. Therefore, it is essential to quantitatively assess the extent to which the targets remain intact under different partitioning strategies.

Specifically, we employ binary annotation maps and apply connected component labeling to delineate all independent foreground regions, each representing a distinct change instance. We then divide each image into four equal parts along the horizontal and vertical axes and examine how the targets are distributed across these resulting patches. A target is considered retained if it is fully contained within a single patch; if it spans multiple patches, we deem it fragmented

$$\text{FFRO} = \frac{N_{\text{retain}}}{N_{\text{total}}}. \tag{9}$$

The fraction of fully retained objects (FFROs) is defined as (9). Here, N_{total} denotes the total number of targets and N_{retain} the number of targets retained intact after partitioning. FFRO quantifies the degree of spatial integrity maintained by the targets postpartitioning: a higher value indicates that the partitioning strategy better preserves continuous target structures.

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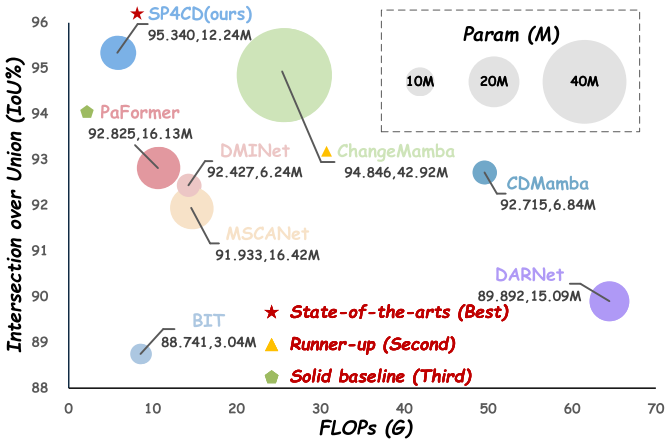


Fig. 1. Comparison of differentmodels on the DSIFN dataset in terms of FLOPs, number of parameters, and performance (IoU%). The bubble size represents the number of parameters. The x-axis and y-axis denote FLOPs (G) and IoU%, respectively.

Algorithm 1 Image to Strip Patches

Input: Input feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$,
Height patch size h_{sp}, Width patch size w_{sp}
Output: Reshaped feature
$\mathbf{Y} \in \mathbb{R}^{(B\cdot h_{sp}\cdot w_{sp})\times C\times (H/h_{sp})\times (W/w_{sp})}$

Reshape $\mathbf{X}$ into patches:
  $\mathbf{X} \leftarrow \text{View}(\mathbf{X}, B, C, H/h_{sp}, h_{sp}, W/w_{sp}, w_{sp})$
Permute dimensions for patch-major layout:
  $\mathbf{X} \leftarrow \mathbf{X}.permute(0, 3, 5, 1, 2, 4)$
Make memory contiguous:
  $\mathbf{X} \leftarrow \text{Contiguous}(\mathbf{X})$
Reshape into batched local patches:
  $\mathbf{Y} \leftarrow \text{View}(\mathbf{X}, -1, C, H/h_{sp}, W/w_{sp})$
return $\mathbf{Y}$

We conducted experiments on the training sets of four CD datasets, and the results are shown in Fig. 2. The results demonstrate that in most cases, a substantial proportion of the targets remain intact after partitioning. This observation provides empirical support for the development of enhanced partitioning strategies and processing modules.

B. Overview

Our proposed network is shown in Fig. 3. Given a pair of remote sensing images pre and postchange, $E_1 \in \mathbb{R}^{3\times H\times W}$ and $E_2 \in \mathbb{R}^{3\times H\times W}$, where dimension 3 denotes the spectral channels, H and W denote the spatial height and width of the images. First, they are fed into the backbone [42] with shared weights, producing features outputs (T_1^a, T_1^b, T_1^c, T_1^d) and (T_2^a, T_2^b, T_2^c, T_2^d). These multiscale change features from each stage are subsequently cropped into vertical and horizontal strip patches. The pseudocode to crop feature maps into strip patches is shown in Algorithm 1. In this article, we set both h_{sp} and w_{sp} to 4 for the purpose of methodological description. The cropped strip patches are used for subsequent cross-temporal spatial interaction and channel interaction. The information obtained after interactions is fused and restored to its precropping size, with pseudocode shown in Algorithm 2.

Algorithm 2 Strip Patches to Image

Input: Windowed feature $\mathbf{Y} \in \mathbb{R}^{(B\cdot h_{sp}\cdot w_{sp})\times C\times H'\times W'}$,
Height patch size h_{sp}, Width patch size w_{sp},
Batch size B
Output: Reconstructed feature map
$\mathbf{X} \in \mathbb{R}^{B\times C\times (H'\cdot h_{sp})\times (W'\cdot w_{sp})}$

Reshape $\mathbf{Y}$ into structured blocks:
  $\mathbf{Y} \leftarrow \text{View}(\mathbf{Y}, B, h_{sp}, w_{sp}, C, H', W')$
Permute to restore spatial layout:
  $\mathbf{Y} \leftarrow \mathbf{Y}.permute(0, 3, 4, 1, 5, 2)$
Make memory contiguous:
  $\mathbf{Y} \leftarrow \text{Contiguous}(\mathbf{Y})$
Reshape back to full spatial dimensions:
  $\mathbf{X} \leftarrow \text{View}(\mathbf{Y}, B, C, H'\cdot h_{sp}, W'\cdot w_{sp})$
return $\mathbf{X}$

These restored feature maps are input as a skip connection into the decoder of the corresponding stage. The fourth stage feature map generated from the backbone (T_1^d, T_2^d) undergoes cropping, interaction, and restoration to serve as input to the decoder. After sequential upsampling, the CD head finally produces a binary change map, which precisely delineates all altered regions within the original input image, allowing accurate localization and identification of changes. The specific decoder block and the segmentation head are shown in the lower left corner of Fig. 3.

C. Patch-Wise Intertemporal Channel Integration

In CD tasks, the model must focus on comparing features across different time phases and further extract and enhance features in the changed regions. During this process, background interference or noise can cause misclassification by the model, and channels typically carry features at different temporal phases. Channel fusion and channel enhancement can effectively address this issue. Based on this, we propose a channel integration method capable of effectively integrating bi-temporal information, termed PICI. In previous operations, we cropped the multilevel features generated by the backbone into strip patches. The data dimension has changed from $T_1 \in \mathbb{R}^{C\times H\times W}$, $T_2 \in \mathbb{R}^{C\times H\times W}$ to $\{T_{1i}\}_{i=1}^4 \in \mathbb{R}^{C\times H/4\times W}$ and $\{T_{2j}\}_{j=1}^4 \in \mathbb{R}^{C\times H/4\times W}$. Patches within the same temporal phase represent different geographic regions in the image. Strip patches of the same color in Fig. 4 represents information from different temporal phases within the same geographic region, such as T_{11} and T_{21}. To highlight the disparity in the same location at different times, we swap the positions of certain strip patches across two phases so that patches representing the same geographic region become adjacent. Subsequently, by establishing feature selection between adjacent strip patches, we achieve information exchange and weighting across channel dimensions. This enables the model to adaptively emphasize channels with prominent structural features while avoiding the interference of information from other geographic regions that would result from applying channel attention across the entire image. If modeling is confined solely to the entire image scale, it can lead to

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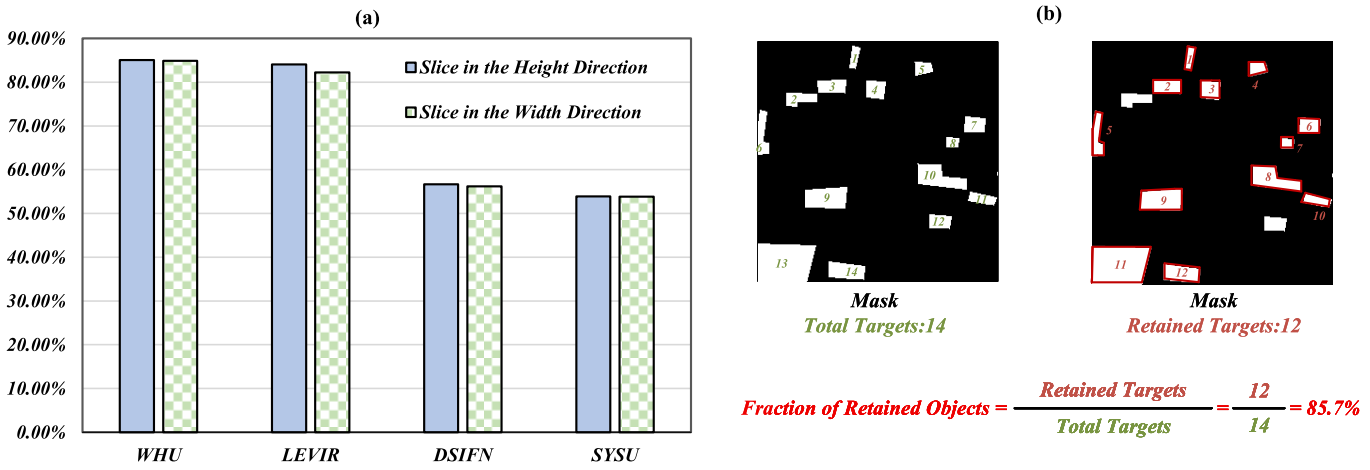


Fig. 2. Motivation for our method. We split the images in the training set into striped panes, count the number of objects in the original image, and count the number of objects that remain in only one pane (fully retained) after the split, in order to calculate the fraction of fully retained objects. There are two slicing ways: in the height and width directions. (a) Fraction of fully retained objects when the four datasets involved in the experiment are split in two directions. (b) Example of a slice in the height direction on the WHU-CD dataset.

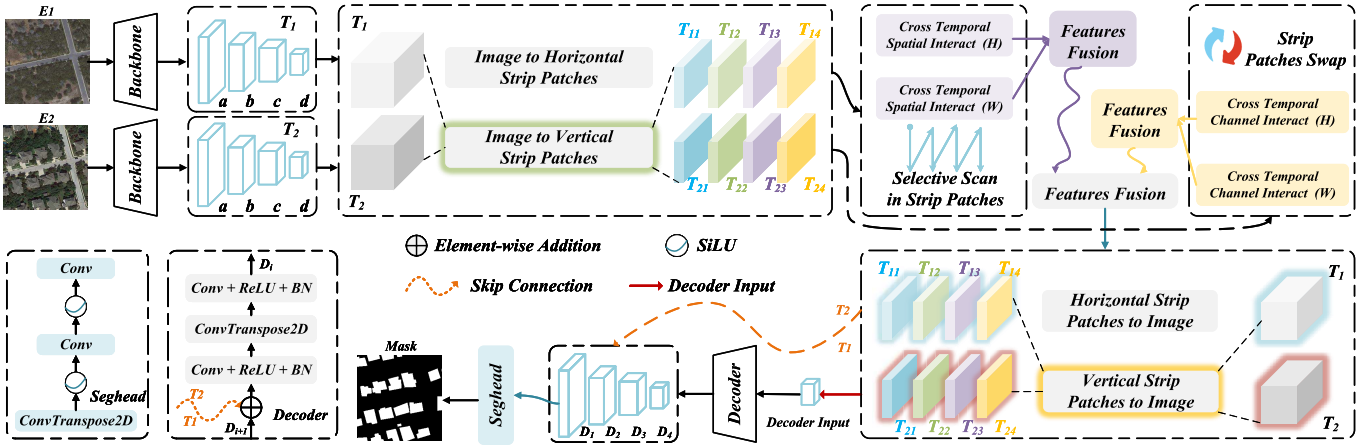


Fig. 3. Overall architecture of the proposed framework SP4CD. It primarily consists of four components: the backbone, cross-temporal spatial interactions, cross-temporal channel interactions, and the decoder.

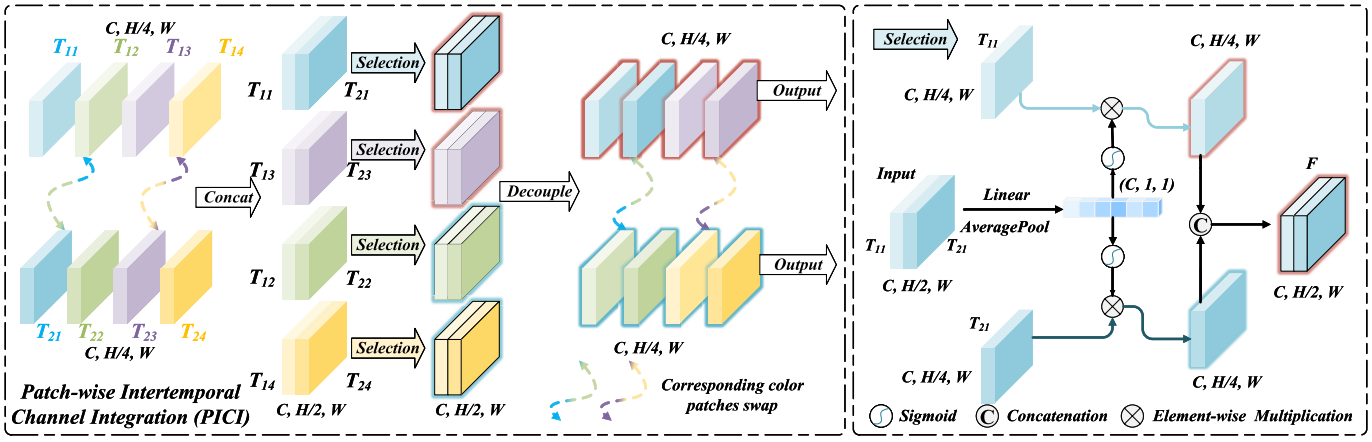


Fig. 4. Architecture of the proposed PICI module. The figure only shows the case of cropping along the H-direction; the same applies to cropping along the W-direction.

insufficient information exchange between channels, resulting in detail loss and redundant feature accumulation.

Specifically, in the feature selection section, we concatenate T_{11} and T_{21} along the H-direction to obtain the Input $\in$

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$\mathbb{R}^{C\times H/2\times W}$, we then apply a *linear layer* to the input to enhance its representational capacity, followed by AveragePool (AP) to obtain a feature map, whose dimension is $(C, 1, 1)$. This feature map, after passing through an activation function, is multiplied as weights onto T_{11} and T_{21}. This enables joint modeling of channel features across different time segments, thereby capturing dynamic dependencies that evolve over time. The above operations can be expressed by the following formula:

$$\begin{aligned} F &= \text{Concat}(T'_{11}, T'_{21}) \\ T'_{11} &= T_{11} \times \sigma(T_A) \\ T'_{21} &= T_{21} \times \sigma(T_A) \\ T_A &= \text{AveragePool}(T_L) \\ T_L &= \text{Linear}(\text{Concat}(T_{11}, T_{21})) \end{aligned} \tag{10}$$

where $\sigma(\cdot)$ indicates the Sigmoid function. Through this operation, the model dynamically allocates attention weights by integrating temporal information, enabling it to emphasize key channels while suppressing irrelevant information. This allows the model to focus more on discriminative features when distinguishing between changes and nonchanges, thereby reducing the probability of false detections. It is worth noting that this is a nearly parameter-free operation.

D. Intertemporal Synergistic Patch Scanning

In CD, many missed detections occur because spatial details in images are not accurately captured by the model. The effectiveness of CD is also affected by lighting variations and noise across different time phases. Therefore, in this article, we design an ISPS method to synchronously model features from two time phases to address these challenges and further enhance features in change regions, as shown in Fig. 5. This method primarily consists of a weight branch and a feature scanning branch. Based on the guidance sequence of the weight branch, we designed two modes: parallel mode and series mode.

1) Weights Based on 3-D Attention: Given the input feature maps T_{11} and $T_{21} \in \mathbb{R}^{C\times H/4\times W}$, we first compute the difference T_D between the two feature maps using absolute value subtraction, then perform feature enhancement via SimAM [43]. The SimAM module computes the attention weights based on the energy of each activation.

Specifically, AP is applied to the resulting difference map T_D to obtain the channel-wise mean value $\mu_{b,c}$. Then, we compute the squared deviation from the channel-wise mean for each spatial pixel, denoted as $E_{b,c,i,j}$, and use the result as the energy for each pixel. This energy is then normalized by the channel-wise variance and a small constant λ, denoted as $Y_{b,c,i,j}$. Finally, we activate the normalized energy and apply it as a weight to the original input feature map through element-wise multiplication, yielding the final attention weights W. The specific steps can be illustrated using the formula below

$$\begin{aligned} W &= \sigma(T'_D) \\ T'_D &= X_{b,c,i,j} \times \sigma(Y_{b,c,i,j}) \\ Y_{b,c,i,j} &= \frac{E_{b,c,i,j}}{4\left(\frac{1}{n}\sum_{i,j} E_{b,c,i,j} + \lambda\right)} + 0.5 \\ E_{b,c,i,j} &= \left(X_{b,c,i,j} - \mu_{b,c}\right)^2 \\ \mu_{b,c} &= \text{AveragePool}(T_D) \\ T_D &= |T_{11} - T_{21}| \end{aligned} \tag{11}$$

where $n = (H/4) \times W - 1$, and λ is typically set to 10^{-4}, $\sigma(\cdot)$ indicates the Sigmoid function, and $X_{b,c,i,j}$ represents each pixel point. The final output weights serve as a gating mechanism to control which features are activated, thereby guiding subsequent feature enhancement branches.

2) Parallel Mode: For the given strip patches T_{11} and $T_{21} \in \mathbb{R}^{C\times H/4\times W}$, we perform spatial feature enhancement on them, respectively. Specifically, the feature maps undergo convolution to achieve positional encoding, followed by linear scanning of the strip patches in four directions [21] to learn global features. Subsequently, layer normalization is employed to standardize features, ensuring consistency and stability in feature distribution across different samples. Subsequently, feature normalization was applied using layer normalization to ensure consistency and stability in feature distribution across different samples, yielding the feature-enhanced results F_1 and $F_2 \in \mathbb{R}^{C\times H/4\times W}$. In parallel mode, these two results are fused as shared features through a Hadamard product with W in (11), achieving synchronous feature fusion to generate the output. Here, W serves as a shared weight to simultaneously model the features of both temporal phases, directly quantifying the consistency and differences between the two temporal images. This aids in filtering out false variations caused by environmental factors. The specific step-by-step process can be represented by the following formula:

$$\begin{aligned} \text{Out} &= W \times F_1 + W \times F_2 \\ F_1 &= \text{LayerNorm}(\text{SS2D}(\text{Conv}(T_{11}))) \\ F_2 &= \text{LayerNorm}(\text{SS2D}(\text{Conv}(T_{21}))). \end{aligned} \tag{12}$$

3) Series Mode: Take T_{11} and T_{21} as examples. T_{11} and T_{21} still perform the spatial feature enhancement mentioned in the parallel mode, yielding outputs F_1 and F_2. Unlike the parallel mode, we first use the output W_1 from (11) (SimAM) as the weight and perform a Hadamard product with F_1. The resulting value then undergoes another iteration of (11) (SimAM) to obtain weight W_2. Finally, we perform a Hadamard product between W_2 and F_2 to obtain the final output. This series of operations can be expressed by the following formula:

$$\begin{aligned} \text{Out} &= W_2 \times F_2 \\ W_2 &= \text{SimAM}(W_1 \times F_1) \\ W_1 &= \text{SimAM}(T_D) \\ F_1 &= \text{LayerNorm}(\text{SS2D}(\text{Conv}(T_{11}))) \\ F_2 &= \text{LayerNorm}(\text{SS2D}(\text{Conv}(T_{21}))). \end{aligned} \tag{13}$$

In this mode, we first use a difference map between two temporal phases to guide the process. This involves extracting the difference regions from the prechange image, which then guides the postchange image. This hierarchical guidance, through gradual fusion and enhancement, better preserves spatial details and contextual relationships. It progressively introduces information from one temporal phase

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onto the other, enabling more refined modeling of cross-temporal dependencies. This facilitates capturing local changes and improving boundary detection.

E. Decoder

After passing through spatial and channel interaction, the feature maps are progressively upsampled in the decoder to generate a segmentation map with the same dimensions as the input image. As shown in Fig. 3, each decoder follows the same structure, consisting of a 1×1 convolution for channel reduction, a 3×3 transposed convolution with a stride of 2 for upsampling, and a final 1×1 convolution for channel restoration. The fourth stage feature map generated from the backbone (T_1^d, T_2^d) undergoes cropping, interaction, and restoration to serve as input to the decoder. Subsequently, each decoder receives the output from the previous decoder along with information from the skip connections. For the segmentation head, we employ a 3×3 transposed convolution with a stride of 2, followed by two 3×3 convolutions to obtain the final results.

IV. Experiment

A. Dataset

We conduct extensive experiments on four public datasets to validate the effectiveness of our proposed network. A brief introduction to these four datasets is provided below.

1) WHU-CD [44]: The dataset is a high-resolution remote sensing dataset tailored for binary building CD tasks. It consists of a pair of aerial images with a spatial resolution of 0.2 m/pixel and a size of $32\,507 \times 15\,354$ pixels, captured in April 2012 and April 2016 in Christchurch, New Zealand. The dataset includes annotations for 12 796 buildings. We divide the images into nonoverlapping 256×256 patches. Then, these were randomly divided into 6096/762/762 training/validation/testing pairs.

2) LEVIR-CD [10]: This is another binary CD dataset that consists of 637 pairs of ultrahigh-resolution (0.5 m/pixel) Google Earth images, each with a spatial size of 1024×1024 pixels. The bi-temporal image pairs span a period of 5–14 years and were collected from 20 urban regions across several cities in Texas, USA. The dataset primarily targets building-related changes, such as construction and demolition, and includes diverse building types, such as villas, high-rise apartments, garages, and large warehouses. We split each image into 256×256 patches without overlap, resulting in 7120/1024/2048 training/validation/testing pairs.

3) DSIFN-CD [12]: The dataset consists of bi-temporal satellite imagery collected from Google Earth, covering six representative cities in China. Each image pair captures structural urban changes over time, such as building construction, demolition, and infrastructure expansion. The data is preprocessed into 256×256 pixel patches, with 14 400/1360/28 training/validation/testing pairs. This dataset provides a valuable benchmark for evaluating CD models on large-scale urban transformations.

4) SYSU-CD [45]: It consists of 20 000 image pairs with a spatial resolution of 0.5 m and a patch size of 256×256 pixels. Captured in Hong Kong between 2007 and 2014, the dataset follows a category-agnostic design, covering a wide range of landscape and infrastructure changes, including urban development, suburban expansion, construction groundwork, vegetation variation, road extension, and coastal reclamation. We adopt the original dataset split and divide it into 12 000/4000/4000 training/validation /testing pairs.

B. Evaluation Metrics

To quantitatively evaluate the performance of the proposed model, we adopt five widely used metrics in the field of CD: overall accuracy (OA), precision (P), recall (R), $F1$ score, and intersection over union (IoU). These metrics comprehensively measure the model's effectiveness in identifying changed regions while balancing different aspects of prediction quality. Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. The definitions of the evaluation metrics are as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

$$F1 = \frac{2}{\mathrm{Recall}^{-1} + \mathrm{Precision}^{-1}} \tag{16}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{17}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{18}$$

The OA represents the ratio of correctly classified pixels, including both changed and unchanged classes, to the total number of pixels. Precision quantifies the percentage of true changed pixels among all pixels predicted as changed. Recall measures the proportion of true changed pixels that are correctly identified by the model. The $F1$ score provides a balanced assessment of the model's accuracy and completeness. IoU measures the spatial intersection between predicted and ground-truth change regions, making it especially relevant for segmentation-based CD. Together, these metrics form a comprehensive evaluation system, enabling reliable assessment of model performance under diverse datasets and conditions.

C. Implementation Details

The proposed model is built on the PyTorch framework with all experiments conducted on an NVIDIA GeForce RTX 4090 GPU. The specific experimental setup includes Ubuntu 20.04, Python 3.10, PyTorch 2.1.1, and CUDA 12.2. We employ the Adam optimizer, the initial learning rate is set to $1e^{-4}$, while β_1 and β_2 are set to 0.9 and 0.99, respectively. The batch size is fixed at six for all datasets. We utilize binary cross-entropy and the Dice coefficient as loss functions [51]

$$L_{\mathrm{total}} = \lambda_1 L_{\mathrm{ce}} + \lambda_2 L_{\mathrm{dice}} \tag{19}$$

$$L_{\mathrm{ce}} = -\frac{1}{N} \sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{20}$$

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TABLE I

ACCURACY ASSESSMENT FOR DIFFERENT BINARY CD MODELS ON WHU-CD AND LEVIR-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND DEEP GREEN

Type	Methods	WHU-CD Pre↑/ Rec↑/ F1↑ / IoU↑ / OA↑ (%)	LEVIR-CD Pre↑ / Rec↑ / F1↑ / IoU↑ / OA↑(%)
CNN-Based	$FC\text{-}EF_{18}$ [13]	95.04 / 94.00 / 94.51 / 90.03 / 99.17	94.87 / 93.66 / 94.25 / 89.58 / 98.90
	$IFNet_{20}$ [12]	97.17 / 94.75 / 95.93 / 92.42 / 99.40	95.52 / 93.83 / 94.66 / 90.26 / 98.99
	$SNUNet_{21}$ [14]	89.03 / 94.61 / 91.61 / 85.45 / 98.63	94.43 / 93.24 / 93.83 / 88.89 / 98.82
	$SEIFNet_{24}$ [11]	96.47 / 95.95 / 96.21 / 92.91 / 99.43	95.26 / 94.09 / 94.67 / 90.27 / 98.98
	$Mobile\text{-}CDNet_{25}$ [46]	91.33 / 92.12 / 91.72 / 85.62 / 98.73	94.17 / 91.86 / 92.98 / 87.53 / 98.68
Transformer-Based	BIT_{21} [33]	94.96 / 96.49 / 95.72 / 92.06 / 99.34	95.22 / 94.18 / 94.69 / 90.31 / 98.99
	$ChangeFormer_{22}$ [34]	95.53 / 94.46 / 94.99 / 90.82 / 99.25	95.24 / 93.48 / 94.34 / 89.73 / 98.93
	$MSCANet_{22}$ [47]	95.63 / 94.96 / 95.29 / 91.34 / 99.29	95.16 / 93.87 / 94.51 / 90.00 / 98.95
	$Paformer_{22}$ [48]	95.95 / 95.17 / 95.66 / 91.96 / 99.34	95.46 / 93.58 / 94.50 / 89.99 / 98.96
	$DARNet_{22}$ [49]	92.65 / 94.56 / 93.58 / 88.52 / 99.00	96.04 / 93.83 / 94.90 / 90.67 / 99.01
	$ACABFNet_{22}$ [35]	96.61 / 93.22 / 94.85 / 90.59 / 99.24	95.13 / 93.26 / 94.17 / 89.45 / 98.90
	$DMINet_{23}$ [50]	96.48 / 96.10 / 96.29 / 93.05 / 99.44	95.31 / 94.09 / 94.69 / 90.31 / 98.99
	$ELGCNet_{24}$ [19]	94.84 / 94.05 / 94.44 / 89.92 / 99.16	94.69 / 92.52 / 93.58 / 88.49 / 98.79
	$ISDANet_{25}$ [20]	90.26 / 96.65 / 93.20 / 87.89 / 98.89	94.82 / 94.35 / 94.59 / 90.13 / 98.96
	$CASP_{25}$ [18]	96.85 / 95.34 / 96.08 / 92.69 / 99.41	95.00 / 94.57 / 94.78 / 90.50 / 99.00
Mamba-Based	$ChangeMamba_{24}$ [39]	96.62 / 94.85 / 95.72 / 92.06 / 99.36	95.25 / 94.16 / 94.70 / 90.32 / 98.99
	$CDMamba_{25}$ [24]	96.05 / 95.27 / 95.66 / 91.95 / 99.34	94.97 / 95.01 / 94.67 / 90.27 / 98.98
	SP4CD-P (Ours)	97.77 / 95.26 / 96.48 / 93.39 / 99.48	95.78 / 94.24 / 95.00 / 90.82 / 99.05
	SP4CD-S (Ours)	97.80 / 94.82 / 96.26 / 93.01 / 99.45	95.53 / 94.05 / 94.78 / 90.45 / 99.01

$$L_{\text{dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \tag{21}$$

where λ_1 and λ_2 are the weighting factors for each loss component. y_i denotes the ground truth label in the ith pixel, while $\hat{y}_i$ represents the predicted probability for that pixel. N refers to the total number of pixels involved in the calculation.

D. Comparison With Existing Methods

To demonstrate the effectiveness of our proposed method, we conduct extensive experiments on the aforementioned four datasets, comparing our model with several classical and novel models. To ensure fairness in comparison, all models are trained under identical experimental conditions and employ the same training strategy. We evaluate the performance of models through evaluation metrics, visualization results, and efficiency analysis.

1) Evaluation Metrics: We employ multiple evaluation metrics to assess the quantitative outcomes of the models. Tables I and II present the metric results for all models on the WUH-CD, LEVIR-CD, DSIFN-CD, and SYSU-CD test sets. The top three results are highlighted in red, blue, and deep green. SP4CD-P and SP4CD-S represent the two models we proposed.

As shown in Table I, our proposed SP4CD-P and SP4CD-S achieved superior results compared to existing CNN-based methods on both the WHU-CD and LEVIR-CD datasets. Although both methods exhibit relatively low Rec values on the WHU-CD dataset, they outperform most approaches on other performance metrics and achieve higher Pre values than other models on LEVIR-CD, indicating exceptional accuracy in detecting change regions. Across both datasets, SP4CD-P delivered the best overall performance, achieving the highest values for $F1$ score, IoU, and OA. Compared to recently proposed Mamba- and Transformer-based approaches (ChangeMamba, CDMamba, ISDANet, and CASP), SP4CD-P achieved the highest $F1$ scores and IoU values, though it was not optimal in precision and recall. On the WHU-CD dataset, SP4CD-P improved $F1$ scores by 0.54%, 0.6%, 3.28%, and 0.4%, respectively, and IoU by 1.33%, 1.44%, 5.5%, and 0.7%, compared to these models.

Table II clearly shows that SP4CD-S achieves the best performance across all metrics on the DSIFN-CD dataset. Regarding the IoU metric, SP4CD-S outperforms the best models based on CNN, Transformer, and Mamba approaches (IFNet, Changformer, and ChangMamba) by 1.35%, 2.45%, and 0.49%, respectively. Although our model shows limited or even inferior performance compared to CDMamba on many metrics of the SYSU-CD dataset, CDMamba consumes approximately eight times more FLOPs (6.05G versus 49.68G). We consider this tradeoff acceptable. Table II presents two datasets with low FFRO rates. The strong performance demonstrated on these datasets indicates that our model can effectively handle datasets with low FFRO rates while simultaneously reducing computational overhead and maintaining model accuracy.

Furthermore, comparing Tables I and II reveals that our two distinct model variants (-P and -S) exhibit strong complementary capabilities across different datasets. They compensate for each other's shortcomings in specific metrics, validating the

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TABLE II

ACCURACY ASSESSMENT FOR DIFFERENT BINARY CD MODELS ON DSIFN-CD AND SYSU-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND DEEP GREEN

Type	Methods	DSIFN-CD Pre↑/ Rec↑/ F1↑ / IoU↑ / OA↑(%)	SYSU-CD Pre↑ / Rec↑ / F1↑ / IoU↑ / OA↑(%)
CNN-Based	FC-EF$_{18}$ [13]	92.50 / 92.32 / 92.42 / 86.30 / 95.73	87.10 / 85.69 / 86.36 / 76.74 / 90.35
	IFNet$_{20}$ [12]	97.25 / 96.48 / 96.86 / 93.99 / 98.24	88.28 / 85.70 / 86.88 / 77.53 / 90.86
	SNUNet$_{21}$ [14]	81.57 / 80.87 / 81.22 / 70.25 / 89.52	88.61 / 78.82 / 82.26 / 71.28 / 88.73
	Mobile-CDNet$_{25}$ [46]	93.00 / 93.50 / 93.25 / 87.67 / 96.16	83.37 / 86.84 / 84.85 / 74.42 / 88.49
Transformer-Based	BIT$_{21}$ [33]	93.52 / 94.25 / 93.88 / 88.74 / 96.52	86.89 / 84.21 / 85.43 / 75.43 / 89.87
	ChangeFormer$_{22}$ [34]	96.12 / 96.40 / 96.26 / 92.89 / 97.88	84.93 / 85.88 / 85.39 / 75.27 / 89.32
	MSCANet$_{22}$ [47]	95.30 / 96.16 / 95.72 / 91.93 / 97.56	85.78 / 83.90 / 84.77 / 74.48 / 89.31
	Paformer$_{22}$ [48]	95.87 / 96.58 / 96.22 / 92.83 / 97.85	87.65 / 84.98 / 86.20 / 76.54 / 90.40
	DARNet$_{22}$ [49]	93.16 / 96.13 / 94.56 / 89.89 / 96.82	85.67 / 84.49 / 85.05 / 74.86 / 89.40
	ACABFNet$_{22}$ [35]	96.43 / 95.94 / 96.18 / 92.76 / 97.86	86.35 / 85.52 / 85.86 / 76.00 / 89.91
	DMINet$_{23}$ [50]	94.91 / 97.18 / 96.00 / 92.43 / 97.69	86.46 / 86.71 / 86.58 / 77.02 / 90.29
	ELGCNet$_{24}$ [19]	94.97 / 96.05 / 95.50 / 91.54 / 97.43	84.69 / 84.85 / 84.77 / 74.41 / 89.00
Mamba-Based	ChangeMamba$_{24}$ [39]	97.75 / 96.91 / 97.32 / 94.85 / 98.50	87.24 / 86.58 / 86.90 / 77.52 / 90.64
	CDMamba$_{25}$ [24]	96.29 / 96.03 / 96.16 / 92.72 / 97.84	86.84 / 87.13 / 87.05 / 77.72 / 90.65
	SP4CD-P (Ours)	97.61 / 97.20 / 97.40 / 94.99 / 98.54	87.74 / 86.15 / 86.91 / 77.54 / 90.76
	SP4CD-S (Ours)	97.94 / 97.25 / 97.59 / 95.34 / 98.65	88.26 / 86.15 / 87.13 / 77.89 / 90.98

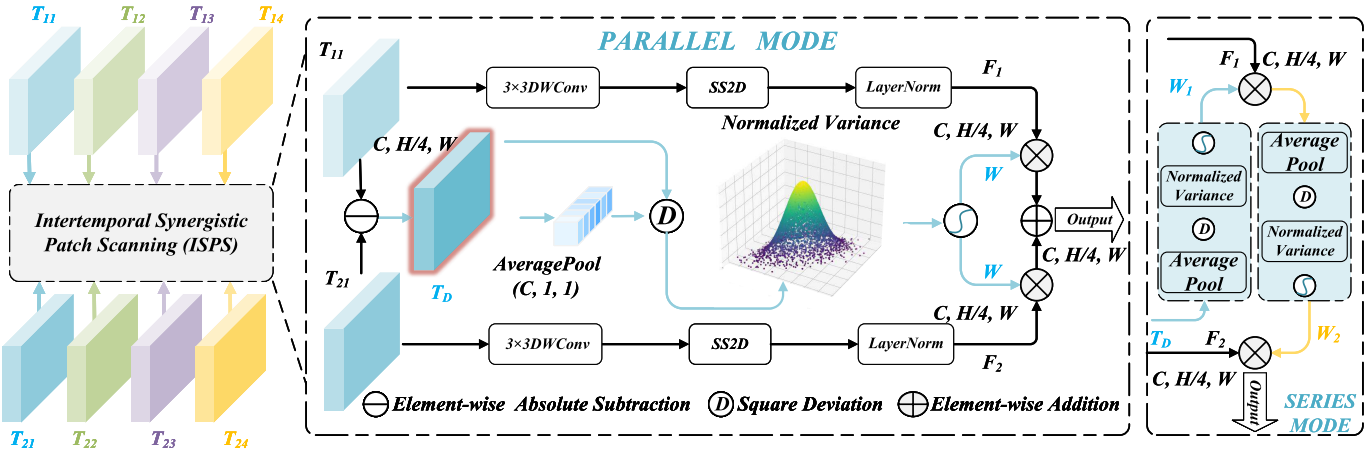


Fig. 5. Architecture of the proposed ISPS module. The figure only shows the case of cropping along the H-direction; the same applies to cropping along the W-direction.

effectiveness and adaptability of our approach across multiple datasets.

2) Visualization Results: To further illustrate the performance distinctions between our proposed model and other models, we conducted visual comparisons with several models (FC-EF, ELGCNet, ACABFNet, CASP, DARNet, Changformer, ChangMamba, and CDMamba) across four datasets. The results are shown in Figs. 6–9. To facilitate interpretation, each detection category is assigned a unique color: TP (white), TN (black), FP (red), and FN (green), indicating correct or erroneous predictions.

Visualization on WHU-CD (Fig. 6): Four representative examples were selected for display in the figure. We selected two examples of small-scale changes (rows 1 and 2) and two instances of large-scale changes (rows 3 and 4). Comparing row 1 reveals that when handling small-scale changes, SP4CD effectively reduces background interference around the change area. The red areas in the second row indicate that SP4CD better captures relationships between changes to lower misclassification rates. The rows 3 and 4 show that SP4CD identifies unchanged areas more accurately, significantly reducing recognition errors. This phenomenon may be due to concentrating feature extraction operations within strip patches, making it more sensitive to minute variations.

Visualization on LEVIR-CD (Fig. 7): We also select four examples on this dataset. The first three rows demonstrate how each model performs when multiple change targets and dispersed change regions are present. From the first row, it is evident that SP4CD significantly outperforms other methods in handling dense CD. In the examples shown in the second row, other methods almost entirely missed the changes, while our SP4CD detection results closely matched the ground truth. The

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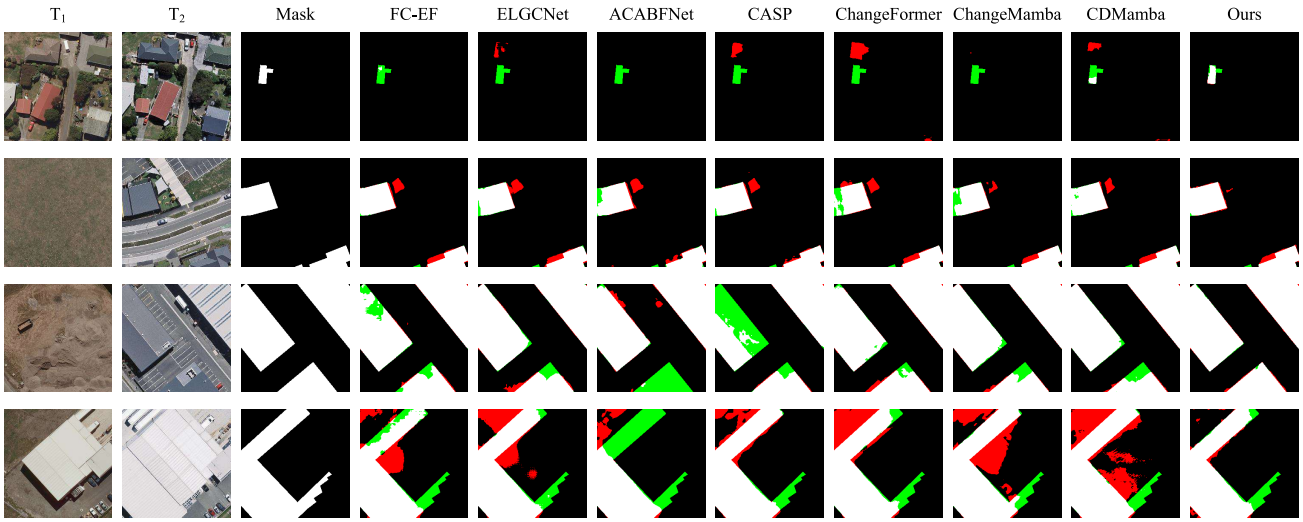


Fig. 6. Visualization results of different methods on the WHU-CD test set. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

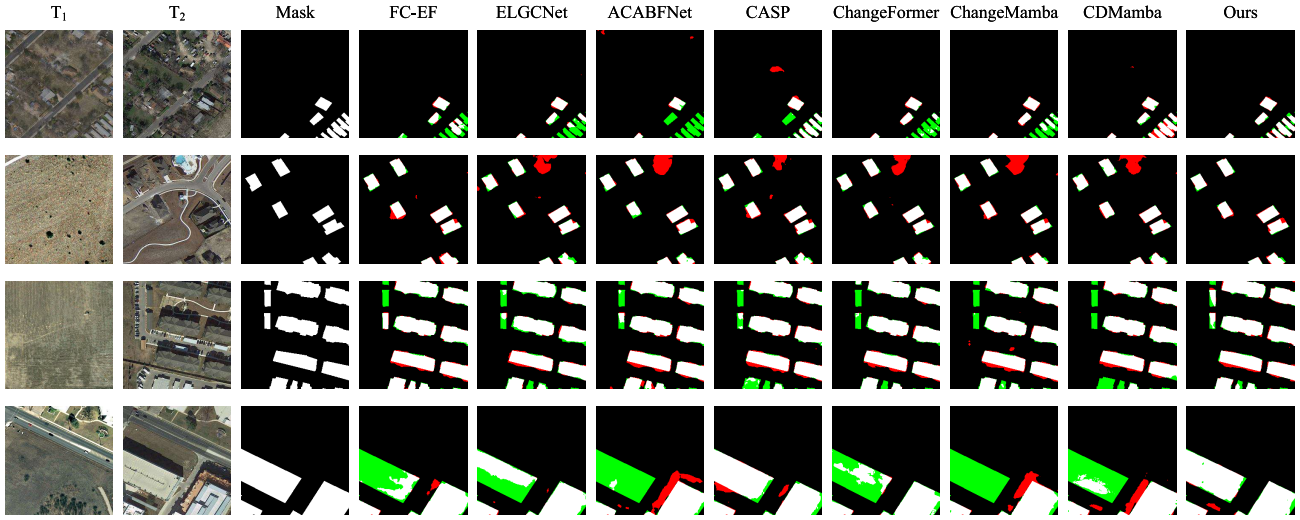


Fig. 7. Visualization results of different methods on the LEVIR-CD test set. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

fourth row depicts a scenario with large-scale changes. Our method maximizes the detection of change regions, further demonstrating its superiority.

Visualization on DSIFN-CD (Fig. 8): In this dataset, we selected examples of both discrete and dense changes. The first two rows demonstrate that SP4CD performs better as the number of change targets increases, particularly when small-scale changes occur. The third and fourth rows indicate that our model excels at distinguishing change areas when their colors closely match the surrounding background, effectively reducing background interference and enabling more accurate edge detection of the change regions.

Visualization on SYSU-CD (Fig. 9): Most changes in the above datasets are regular. For the SYSU-CD dataset, we selected cases with irregular change areas. In the first two scenarios, seasonal variations affected segmentation. The distinct characteristics of vegetation across seasons became interference factors for CD. Our proposed model effectively mitigates this impact, whereas other models mistakenly identified it as a change. When large-scale changes occur in irregularly shaped areas, SP4CD can also achieve detection tasks to a significant extent. This further validates that our model is not only effective for multi-object changes but can also handle large-scale changes.

3) Efficiency Analysis: We present a comparison of SP4CD with several classical methods in terms of parameters and FLOPs on the DSIFN dataset. These methods include ChangeMamba, CDMamba, BIT, PaFormer, DMINet, MSCANet, and DARNet. The results are shown in Fig. 1. Different colored circles represent different models, with circle diameter indicating the model's parameter count. The horizontal and vertical axes correspond to the model's FLOPs and IoU, respectively. Thus, models represented by circles closer to the upper-left corner demonstrate superior performance. It can be observed that our

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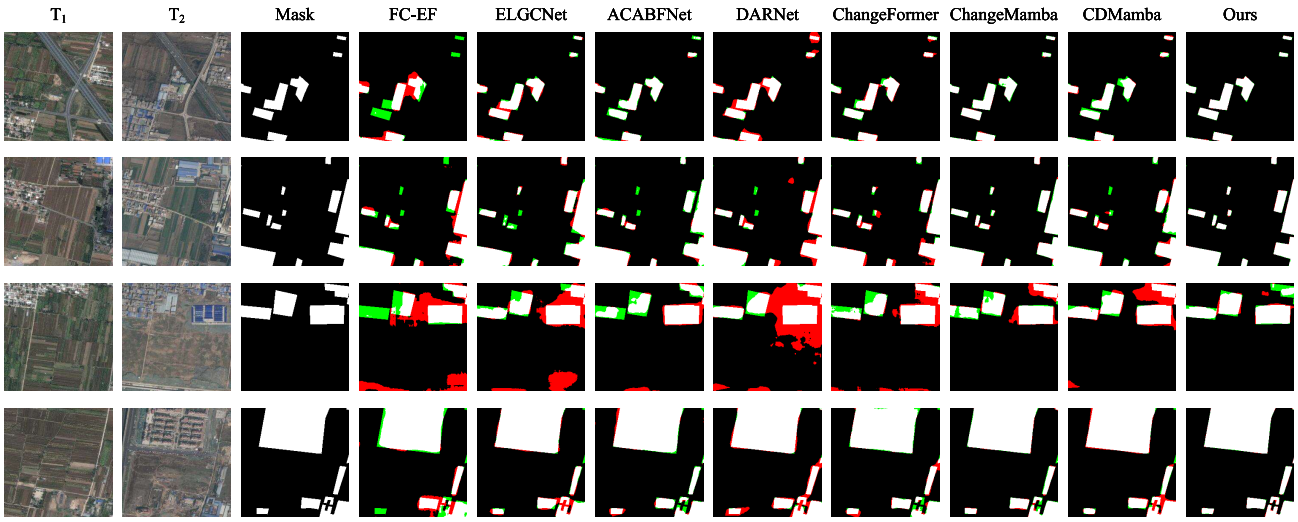


Fig. 8. Visualization results of different methods on the DSIFN-CD test set. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

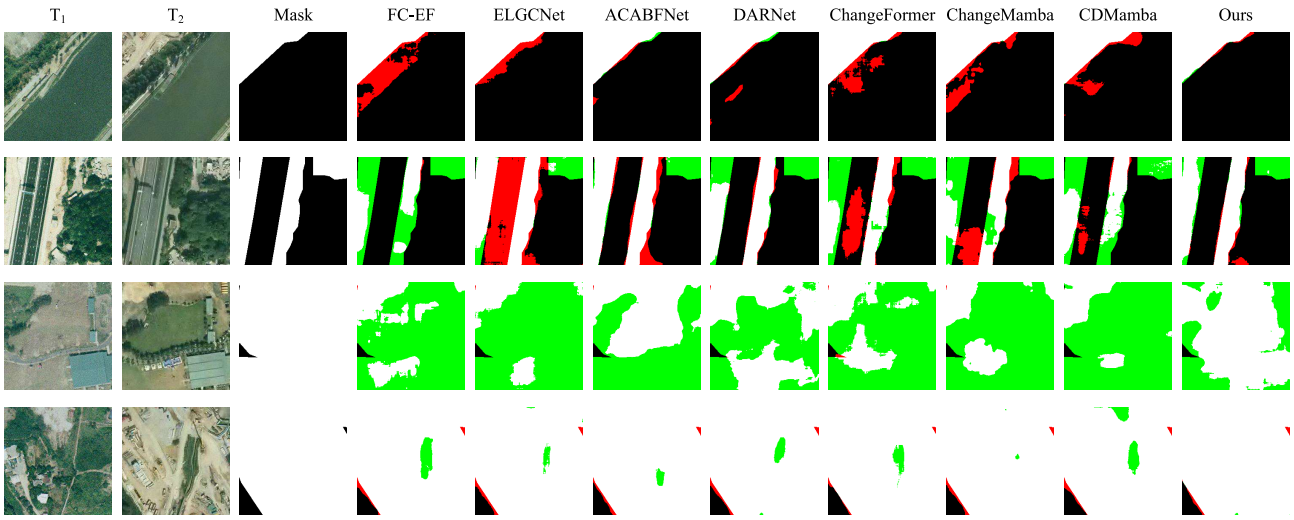


Fig. 9. Visualization results of different methods on the SYSU-CD test set. White represents a true positive, black is a true negative, red indicates a false positive, and green stands as a false negative.

proposed SP4CD achieves the best performance on this dataset with minimal FLOPs. For example, compared to CDMamba, we achieved a reduction of 87. 82% in FLOPs while improving the IoU metric by 2.62%. This advantage stems from our designed PICI and ISPS modules. The PICI module achieves channel enhancement by swapping patches from different temporal phases. This operation incurs minimal parameter overhead, as only the linear layer involves a small number of parameters, ensuring lightweight processing. The ISPS module reduces the computational burden of full-image scanning by implementing a selective scan within strip patches. In addition, our backbone achieves efficient feature extraction with minimal computation and overhead, further contributing to our model's lightweight nature. Therefore, the FLOPs of SP4CD have been reduced by 76%–96% compared to methods, such as ChangeMamba, ChangeFormer, and ELGCNet, offering new insights for resource-constrained applications.

E. Ablation Studies

Ablation studies were carried out on the WHU-CD and LEVIR-CD datasets to quantitatively assess the role of each architectural module. We also provide an in-depth discussion of key hyperparameter choices, offering insights into their influence on model behavior and performance. It is worth noting that all numerical values in the ablation study results table are expressed as percentages.

1) Impact of Individual Component in SP4CD: To validate the effectiveness of each key component in our network, we design multiple sets of ablation studies on the WHU-CD and LEVIR-CD datasets. The results of the ablation experiments on these two datasets are shown in Tables III and IV, respectively. The tables reveal that adding either the ISPS or PICI module to the baseline consistently yields improved experimental results. When both modules are integrated, the critical metrics for binary CD, $F1$ score, and IoU

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TABLE III

ABLATION STUDY ON WHU-CD DATASET WITH DIFFERENT COMPONENTS. THE BEST RESULTS ARE BOLDED

Components	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
Baseline	91.33	90.62	90.97	84.51	98.64
w/ *ISPS-P*	**97.80**	94.50	96.08	92.69	99.42
w/ *ISPS-S*	97.45	95.04	96.21	92.92	99.44
w/ *PICI*	97.52	94.93	96.19	92.88	99.43
w/ *ISPS-P + PICI*	97.77	**95.26**	**96.48**	**93.39**	**99.48**
w/ *ISPS-S + PICI*	**97.80**	94.82	96.26	93.01	99.45

TABLE IV

ABLATION STUDY ON LEVIR-CD DATASET WITH DIFFERENT COMPONENTS. THE BEST RESULTS ARE BOLDED

Components	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
Baseline	92.97	91.98	92.47	86.74	98.56
w/ *ISPS-P*	95.67	93.93	94.80	90.46	99.01
w/ *ISPS-S*	95.53	93.86	94.68	90.28	98.99
w/ *PICI*	**95.97**	93.49	94.76	90.43	99.01
w/ *ISPS-P + PICI*	95.78	**94.24**	**95.00**	**90.82**	**99.05**
w/ *ISPS-S + PICI*	95.53	94.05	94.78	90.45	99.01

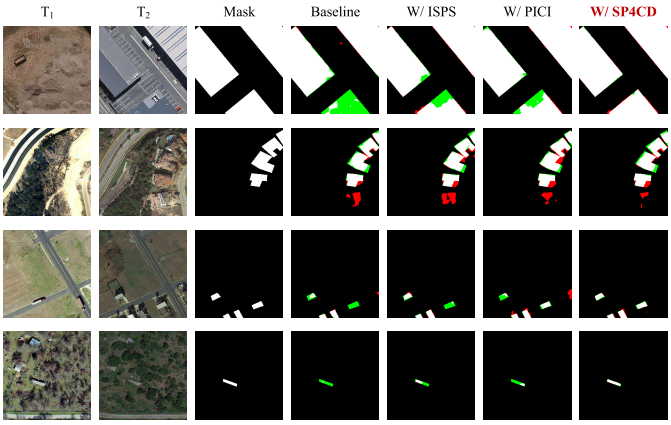


Fig. 10. Detection results of variations in the same input across different components. Row 1 originates from the WHU-CD dataset, while rows 2–4 originate from the LEVIR-CD dataset.

achieve the greatest gains on the two datasets. We believe such improvements can quantitatively demonstrate the effectiveness of the two proposed models. Simultaneously, aggregating spatial and channel features can synergistically enhance model performance.

To further visually validate the effectiveness of each component, several illustrative examples are presented in Fig. 10. These demonstrate the distinct effects of each module during the detection task and their progressive outcomes. The first row illustrates that both ISPS and PICI can capture more change regions compared with the baseline. When integrated into the model, SP4CD achieves nearly the most accurate predictions. The second row reveals that ISPS and PICI may occasionally produce more false detections than the baseline; however, their combination significantly reduces such errors. The third row demonstrates that when multiple change regions exist, our proposed modules individually detect specific regions, while their collaboration enables the identification of all change areas. The fourth row shows that the two modules focus on different parts of a single change region, and their integration leads to accurate detection.

Furthermore, we examined the stage-wise attention regions of the two modules through heatmap visualization [52]. As illustrated in Fig. 11, representative examples from the DSIFN-CD and SYSU-CD datasets are provided to highlight their distinct focus patterns. The heatmap visualizations further demonstrate that the two proposed modules exhibit notable complementarity in feature attention. Their synergistic integration effectively enhances both the accuracy and robustness of change region detection.

TABLE V

ABLATION STUDY ON WHU-CD WITH DIFFERENT PATCH SIZES. THE BEST RESULTS ARE BOLDED

Patch Sizes	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(1,1,1,1)	97.74	94.85	96.24	92.97	99.45
(16,8,4,2)	97.68	94.50	96.03	92.61	99.42
(8,8,8,8)	97.40	95.03	96.18	92.86	99.43
(4,4,4,4)	**97.77**	**95.26**	**96.48**	**93.39**	**99.48**
(4,4,2,1)	95.62	90.61	92.96	87.54	98.99

TABLE VI

ABLATION STUDY ON LEVIR-CD WITH DIFFERENT PATCH SIZES. THE BEST RESULTS ARE BOLDED

Patch Sizes	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(1,1,1,1)	95.70	93.91	94.79	90.47	99.01
(16,8,4,2)	95.52	93.98	94.74	90.39	99.00
(8,8,8,8)	95.74	93.65	94.67	90.27	98.99
(4,4,4,4)	**95.78**	94.24	**95.00**	**90.82**	**99.05**
(4,4,2,1)	93.02	**94.65**	93.82	88.87	98.78

2) Impact of Different Patch Sizes: We conducted multiple design experiments on patch sizes for image cropping at different encoder stages. Considering the encoder's progressive downsampling of images, which alters image dimensions, we designed our approach such that the patch size for the next stage does not exceed that of the previous stage. Experimental results on the WHU-CD and LEVIR-CD datasets are shown in Tables V and VI.

Here, (a, b, c, d) represents the number of cropped segments for the corresponding images at each of the four encoder stages. For example, (16, 8, 4, 2) indicates that the image is cropped into 16, 8, 4, and 2 segments horizontally and vertically, respectively, across the four encoding stages. In particular, (1, 1, 1, 1) means that no cropping is performed at any encoding stage, which means that subsequent feature interactions occur directly on the original image. The ablation study in this section employed the proposed SP4CD-P. From the two tables, it can be observed that even without image cropping, our module's performance still surpasses some mainstream

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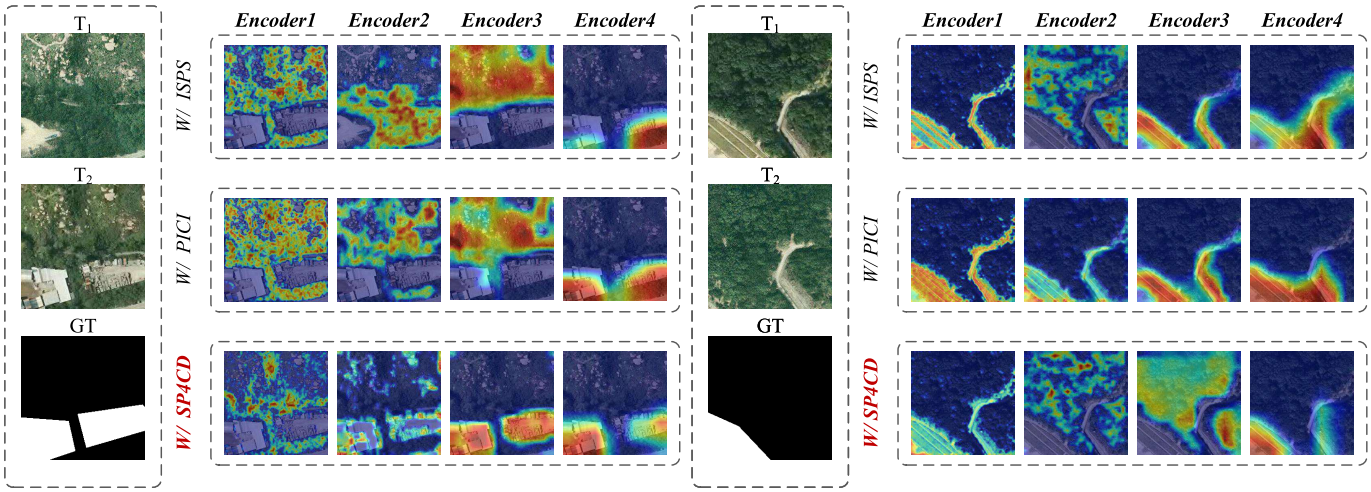


Fig. 11. Visualization results of the differential feature maps on DSIFN-CD and SYSU-CD at different stages. Blue indicates the strongest negative impact on classification, while red indicates the strongest positive impact.

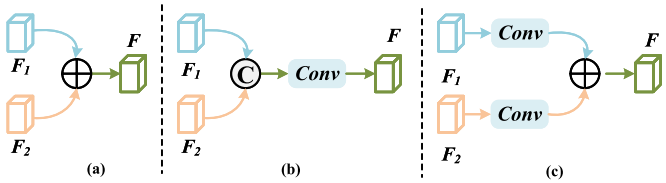


Fig. 12. Different features fusion strategies. F_1 and F_2 represent different features, and F represents the result after feature fusion. (a)–(c) represent three feature fusion strategies.

TABLE VII

ABLATION STUDY ON WHU-CD WITH DIFFERENT FEATURES FUSION STRATEGIES. THE BEST RESULTS ARE BOLDED

Mode	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(a)	97.12	**95.34**	96.21	92.92	99.43
(b)	97.48	94.78	96.08	92.70	99.42
(c)	**97.77**	95.26	**96.48**	**93.39**	**99.48**

models, further validating the effectiveness of the two modules we designed. Comparing the metric results between the (4, 4, 4, 4) and (1, 1, 1, 1) configurations reveals improvements across all performance indicators on both datasets. The IoU increased by 0.42% and 0.35%, respectively. This demonstrates that by partitioning the image, we can more accurately capture the relationships between change targets while ensuring the vast majority remain within the patch. Concentrating feature extraction and long sequence modeling within patches significantly enhances detection performance. The other two experimental designs employed larger patch segmentation sizes, resulting in more image fragments. This approach risks splitting a complete change target across one or more patches, disrupting its original geometric information and hindering change target extraction and detection. Consequently, these designs yield inferior performance metrics compared to processing the original image directly.

TABLE VIII

ABLATION STUDY ON LEVIR-CD WITH DIFFERENT FEATURES FUSION STRATEGIES. THE BEST RESULTS ARE BOLDED

Mode	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(a)	95.34	93.94	94.75	90.41	99.00
(b)	95.47	**96.37**	94.91	90.68	99.03
(c)	**95.78**	94.24	**95.00**	**90.82**	**99.05**

3) Impact of Different Features Fusion Strategies: As shown in Fig. 3, we fuse features in both horizontal and vertical directions. We also fuse features after spatial and channel interactions. We discuss the features and fusion strategies involved here. The ablation study in this section employed the proposed SP4CD-P. As illustrated in Fig. 12, we propose three feature fusion strategies: (a)–(c). Their performance on the WHU-CD and LEVIR-CD datasets is summarized in Tables VII and VIII. The experimental results demonstrate that the fusion strategy applied after enhancing the multisource features F_1 and F_2 consistently outperforms the other two approaches across all key evaluation metrics. This indicates that performing feature enhancement prior to fusion allows the network to capture more discriminative and complementary information from each source, thereby yielding a more robust representation. Consequently, strategy (c) is selected in this study as the optimal feature fusion scheme to achieve superior performance and ensure reliable CD results.

4) Impact of Coefficients in the Loss Function: Since the loss function employed in this article combines cross-entropy and dice score, we conducted experiments on the LEVIR-CD dataset to validate the effectiveness of different loss function coefficient combinations. Bar charts were used to visualize the Pre, Rec, $F1$ scores, and IoU values, with the experimental results shown in Fig. 13. We designed four loss function coefficient combinations: [0.5, 0.5], [1, 0], [0, 1], [1, 0.5], and [0.5, 1]. The numbers in parentheses represent the cross-entropy loss coefficient and the dice loss coefficient, respectively.

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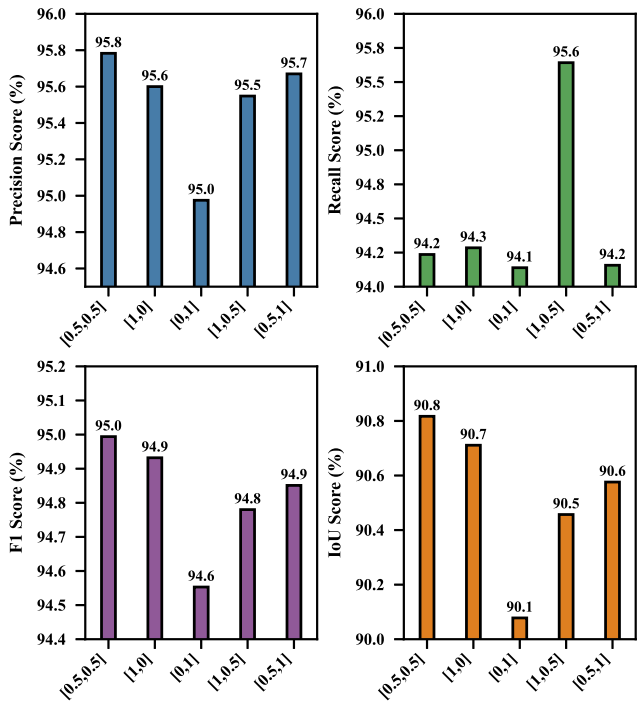


Fig. 13. Ablation study on different coefficients of the loss function on LEVIR-CD, where the bars indicate the $F1$ score, and [0.5, 0.5], [1, 0], and [0, 1] denote the coefficients of the cross-entropy and dice loss functions, respectively.

V. Conclusion

In this article, we introduce a novel perspective on CD tasks and propose a lightweight network: SP4CD. Specifically, after analyzing the characteristics of multiple datasets and performing statistical analysis, we propose processing within strip patches. To fuse temporal information and reduce the model's false alarm rate, we introduce the PICI module. Through the exchange, concatenation, selection, and decoupling of strip patches, we achieve the enhancement of the low-parameter channel information. In addition, we design the ISPS module to capture more precise spatial information. Employing difference-guided scanning for long-sequence modeling, it captures spatial features across dual time phases, enhancing the model's CD accuracy while reducing false negatives. The results on four public datasets demonstrate that our method performs similar to previous work. Furthermore, we performed ablation experiments on SP4CD components and other hyperparameters to validate their effectiveness. Although multiple experiments on strip patch sizes are performed in this article, limitations remain. Future work will explore adaptive rectangular patches as alternatives to the predefined strip patches and extend them to multiclass CD and SAR data, allowing better preservation of structural integrity and enhanced robustness under lightweight conditions.

References

[1] Y. Afaq and A. Manocha, "Analysis on change detection techniques for remote sensing applications: A review," *Ecological Informat.*, vol. 63, Jul. 2021, Art. no. 101310.

[2] W. Xiao et al., "Measuring three-dimension urban expansion using multi-source data and change detection algorithm: A case study of Shanghai," *Cities*, vol. 158, Mar. 2025, Art. no. 105682.

[3] M. Kucharczyk and C. H. Hugenholtz, "Remote sensing of natural hazard-related disasters with small drones: Global trends, biases, and research opportunities," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112577.

[4] V. Pushpalatha, P. B. Mallikarjuna, H. N. Mahendra, S. R. Subramoniam, and S. Mallikarjunaswamy, "Land use and land cover classification for change detection studies using convolutional neural network," *Appl. Comput. Geosci.*, vol. 25, Feb. 2025, Art. no. 100227.

[5] P. de Bem, O. de Carvalho Junior, R. F. Guimarães, and R. T. Gomes, "Change detection of deforestation in the Brazilian Amazon using Landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020.

[6] C. Huo, K. Chen, K. Ding, Z. Zhou, and C. Pan, "Learning relationship for very high resolution image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3384–3394, Aug. 2016.

[7] D. K. Seo, Y. H. Kim, Y. D. Eo, M. H. Lee, and W. Y. Park, "Fusion of SAR and multispectral images using random forest regression for change detection," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 10, p. 401, Oct. 2018.

[8] R. Shang, L. Qi, L. Jiao, R. Stolkin, and Y. Li, "Change detection in SAR images by artificial immune multi-objective clustering," *Eng. Appl. Artif. Intell.*, vol. 31, pp. 53–67, May 2014.

[9] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," *Remote Sens.*, vol. 16, no. 13, p. 2355, Jun. 2024.

[10] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[11] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

[12] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[13] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[14] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[16] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[17] Z. Yang, W. Zhang, Q. Li, W. Ni, J. Wu, and Q. Wang, "C^{2}Net: Road extraction via context perception and cross spatial-scale feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024.

[18] Q. Wang, M. Zhang, J. Ren, and Q. Li, "Exploring context alignment and structure perception for building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–10, 2025.

[19] M. Noman, M. Fiaz, H. Cholakkal, S. Khan, and F. S. Khan, "ELGC-Net: Efficient local–global context aggregation for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024.

[20] H. Ren, M. Xia, L. Weng, H. Lin, J. Huang, and K. Hu, "Interactive and supervised dual-mode attention network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–18, 2025.

[21] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[22] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[23] J. Jiao et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.

[24] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–16, 2025.

[25] Y. Xing, Y. Jia, S. Gao, J. Hu, and R. Huang, "Frequency-enhanced mamba for remote sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

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[26] J. Li, M. Xu, H. Chen, W. Liu, L. Chen, and Y. Xie, "Spatio-temporal pruning for training ultra-low-latency spiking neural networks in remote sensing scene classification," *Remote Sens.*, vol. 16, no. 17, p. 3200, Aug. 2024.

[27] D. Duan, P. Liu, B. Hui, and F. Wen, "Brain-inspired online adaptation for remote sensing with spiking neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–18, 2025.

[28] S. Yang, B. Linares-Barranco, Y. Wu, and B. Chen, "Self-supervised high-order information bottleneck learning of spiking neural network for robust event-based optical flow estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2280–2297, Apr. 2025.

[29] F. Rahman, B. Vasu, J. V. Cor, J. Kerekes, and A. Savakis, "Siamese network with multi-level features for patch-based change detection in satellite imagery," in *Proc. IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, Nov. 2018, pp. 958–962.

[30] L. Xue, X. Wang, Z. Wang, G. Li, H. Song, and Z. Song, "A lightweight patch-level change detection network based on multilayer feature compression and sensitivity-guided network pruning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–19, 2024.

[31] H. Chen, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 87–105, Dec. 2023.

[32] T. Lei et al., "Ultralightweight spatial–spectral feature cooperation network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.

[33] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[34] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[35] L. Song, M. Xia, L. Weng, H. Lin, M. Qian, and B. Chen, "Axial cross attention meets CNN: Bibranch fusion network for change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 21–32, 2023.

[36] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[37] J. Zhou, Y. Liu, B. Peng, L. Liu, and X. Li, "MaDiNet: Mamba diffusion network for SAR target detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 11, pp. 10787–10800, Nov. 2025.

[38] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

[39] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–20, 2024.

[40] Z. Wu et al., "CD-lamba: Boosting remote sensing change detection via a cross-temporal locally adaptive state space model," 2025, *arXiv:2501.15455*.

[41] S. Liu et al., "CD-STMamba: Toward remote sensing image change detection with spatio-temporal interaction mamba model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10471–10485, 2025.

[42] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.

[43] L. Yang, R. Zhang, L. Li, and X. Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proc. 38th Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 11863–11874.

[44] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[45] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[46] Y. Fu, H. Peng, T. Zhao, Y. Li, J. Peng, and D. Zhang, "Lightweight remote sensing change detection with progressive multi scale difference aggregation," *Sci. Rep.*, vol. 15, no. 1, p. 30203, Aug. 2025.

[47] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[48] M. Liu, Q. Shi, Z. Chai, and J. Li, "PA-former: Learning prior-aware transformer for remote sensing building change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[50] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.

[51] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. J. Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, 2017, pp. 240–248.

[52] M. B. Muhammad and M. Yeasin, "Eigen-CAM: Class activation map using principal components," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–7.

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